01/04/2025

CODE-A

EDITION 2025 EDITION

Aakash

Medical | IIT-JEE | Foundations

Corporate Office**:** AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

720

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

PST-3

Time : 180 Mins.

Answers

1.	(4)	37.	(1)	73.	(1)	109.	(2)	145.	(2)
2.	(4)	38.	(3)	74.	(2)	110.	(3)	146.	(1)
3.	(2)	39.	(4)	75.	(4)	111.	(3)	147.	(4)
4.	(1)	40.	(4)	76.	(4)	112.	(3)	148.	(4)
5.	(2)	41.	(2)	77.	(4)	113.	(4)	149.	(1)
6.	(1)	42.	(3)	78.	(1)	114.	(2)	150.	(2)
7.	(1)	43.	(4)	79.	(4)	115.	(4)	151.	(2)
8.	(2)	44.	(3)	80.	(3)	116.	(4)	152.	(3)
9.	(2)	45.	(3)	81.	(3)	117.	(3)	153.	(1)
10.	(3)	46.	(2)	82.	(2)	118.	(2)	154.	(4)
11.	(3)	47.	(3)	83.	(2)	119.	(3)	155.	(4)
12.	(1)	48.	(1)	84.	(1)	120.	(3)	156.	(2)
13.	(3)	49.	(3)	85.	(3)	121.	(2)	157.	(3)
14.	(3)	50.	(4)	86.	(4)	122.	(2)	158.	(3)
15.	(1)	51.	(2)	87.	(2)	123.	(2)	159.	(2)
16.	(3)	52.	(1)	88.	(3)	124.	(1)	160.	(1)
17.	(3)	53.	(1)	89.	(1)	125.	(4)	161.	(4)
18.	(3)	54.	(3)	90.	(1)	126.	(1)	162.	(2)
19.	(2)	55.	(4)	91.	(3)	127.	(2)	163.	(2)
20.	(1)	56.	(2)	92.	(3)	128.	(3)	164.	(4)
21.	(1)	57.	(4)	93.	(3)	129.	(1)	165.	(2)
22.	(2)	58.	(1)	94.	(4)	130.	(2)	166.	(1)
23.	(4)	59.	(3)	95.	(2)	131.	(2)	167.	(2)
24.	(3)	60.	(2)	96.	(3)	132.	(3)	168.	(3)
25.	(1)	61.	(1)	97.	(1)	133.	(3)	169.	(1)
26.	(2)	62.	(2)	98.	(4)	134.	(1)	170.	(3)
27.	(3)	63.	(1)	99.	(2)	135.	(2)	171.	(2)
28.	(1)	64.	(4)	100.	(3)	136.	(4)	172.	(2)
29.	(4)	65.	(3)	101.	(2)	137.	(4)	173.	(1)
30.	(3)	66.	(2)	102.	(2)	138.	(1)	174.	(1)
31.	(3)	67.	(4)	103.	(3)	139.	(4)	175.	(2)
32.	(2)	68.	(3)	104.	(1)	140.	(4)	176.	(2)
33.	(4)	69.	(2)	105.	(2)	141.	(4)	177.	(1)
34.	(2)	70.	(4)	106.	(3)	142.	(1)	178.	(3)
35.	(3)	71.	(1)	107.	(4)	143.	(1)	179.	(4)
36.	(1)	72.	(3)	108.	(2)	144.	(1)	180.	(4)

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
Relative error is dimensionless

(2) **Answer :** (4)
Solution:
If a measured quantity has *n* significant figures, then reliable digits in if are (n – 1)

(3) **Answer :** (2)
Solution:
$E = h\nu$
$ML^2T^{-2} = h \times \frac{1}{T}$
$h = ML^2T^{-1}$
ML^2T^{-1} is dimension of angular impulse

(4) **Answer :** (1)
Solution:
Correctness of a equation is verified by the principle of homogeneity and all unitless quantities are is dimensionless

(5) **Answer :** (2)
Solution:
$F = l \times b$
$\frac{\Delta A}{A} \times 100 = \frac{\Delta l}{l} \times 100 + \frac{\Delta b}{b} \times 100$
$= \frac{0.02}{20} \times 100 + \frac{0.02}{10} \times 100$
= 0.02 × 5 + 0.02 × 10
= 0.1 + 0.2 = 0.3%

(6) **Answer :** (1)
Solution:
$p = at + bt^2$
$at = MLT^{-1}$
$a = MLT^{-2}$
$bt^2 = MLT^{-1}$
$b = MLT^{-3}$

(7) **Answer :** (1)
Solution:
$n_1U_1 = n_2U_2$
$100\left[M_1L_1^2T_1^{-3}\right] = n_2\left[M_2L_2^2T_2^{-3}\right]$
$n_2 = 100\left[\frac{M_1}{M_2} \times \frac{L_1^2}{L_2^2} \times \left(\frac{T_2}{T_1}\right)^3\right]$
$= 100\left[\frac{M_1}{\frac{M_1}{2}} \times \frac{L_1^2}{\left(\frac{L_1}{2}\right)^2} \times \left[\frac{\frac{T_1}{2}}{T_1}\right]\right]^3$
$= 100 \times 2 \times 4 \times \frac{1}{8}$ = 100 W

(8) **Answer :** (2)
Solution:
$(\text{Energy})^n = (\text{Force})^{\frac{1}{2}} (\text{Area})^{\frac{1}{4}}$
$= (\text{Force})^{\frac{1}{2}} (\text{length})^{\frac{1}{2}}$
$= (\text{Work})^{\frac{1}{2}}$

$n = \frac{1}{2}$

(9) **Answer :** (2)
Solution:
$S = ut + \frac{1}{2}at^2$
$\frac{h}{x} = \frac{1}{2}gt^2$
$h = \frac{1}{2}gT^2$
$\frac{1}{x} = \frac{t^2}{T^2} \Rightarrow T = t\sqrt{x}$

(10) **Answer :** (3)
Solution:
Velocity is given by slope of position time curve.

(11) **Answer :** (3)
Solution:
$a = \frac{dv}{dt}$ slope of velocity – time graph is acceleration

(12) **Answer :** (1)
Solution:
$v_{\text{avg}} = \frac{v_1 + v_2}{2}$
$v_1 = 0,\ v_2 = \sqrt{gh}$
$v_{\text{avg}} = \frac{\sqrt{gh}}{2}$

(13) **Answer :** (3)
Solution:
Acceleration opposite to the velocity slowing down, distance travelled by a body is always positive.

(14) **Answer :** (3)
Solution:
$V_{\text{avg}} = \frac{\frac{1}{2}\times20\times4 + \left(\frac{1}{2}\times20\times2\right)\times2}{8}$
$= \frac{40+40}{8} = \frac{80}{8} = 10$ m/s

(15) **Answer :** (1)
Solution:
$v = 2t + 3t^2$
$\frac{dx}{dt} = 2t + 3t^2$
$\int dx = \int_2^3 \left(2t + 3t^2\right) dt$
$x = \left[t^2\right]_2^3 + \left[t^3\right]_2^3 = 5 + 19 = 24$ m

(16) **Answer :** (3)
Solution:
$t = \frac{d_1 + d_2}{v_1 + v_2}$
$10 = \frac{200 + x}{25}$
$\Rightarrow x = 50$ m

(17) **Answer :** (3)
Solution:
$\overrightarrow{v_{pg}} = ui$
$\overrightarrow{v_{pg}} = u\hat{i} + gt\,\hat{j}$
$\overrightarrow{v_{op}} = u\hat{i} - gt\,\hat{j} - ui = -gt\,\hat{j}$
$\left|\overrightarrow{v_{og}}\right| = \sqrt{u^2 + (gt)^2}$

(18) **Answer :** (3)

Solution:
$P_i = mv = P$
$P_t = mv\cos 60°$
$= \frac{mv}{2} = \frac{P}{2}$

(19) Answer : (2)
Solution:
$\left|\overrightarrow{R}\right| = A^2 + B^2$

$R^2 = B^2 + \frac{B^2}{4}$

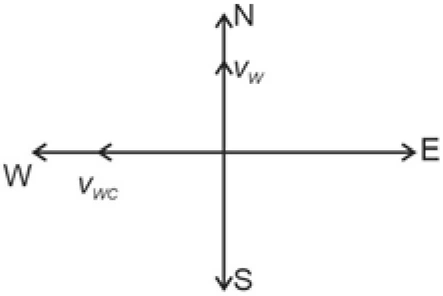
$R^2 = \frac{5B^2}{4}$

$R = \frac{\sqrt{5}B}{2}$

$\cos\theta = \frac{B}{R}$

$\cos\theta = \frac{2}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(20) Answer : (1)
Solution:



$\overrightarrow{v_{wc}} = \overrightarrow{v_{wg}} - \overrightarrow{v_{cg}}$
$\overrightarrow{v_{cg}} = \overrightarrow{v_{wg}} - \overrightarrow{v_{wc}}$
$= 5\hat{j} - \left(-5\hat{i}\right) = 5\hat{j} + 5\hat{i} = 5\sqrt{2}\left(N - E\right)$

(21) Answer : (1)
Solution:
$\overrightarrow{v} = u\cos\theta\hat{i} + \left(4\sin\theta - gt\right)\hat{j}$

$= 80\cos 30\hat{i} + \left(80\sin 30^\circ - 10\right)\hat{j}$

$= \left(40\sqrt{3}\hat{i} + 30\hat{j}\right)$ m/s

(22) Answer : (2)
Solution:
$H_t = \frac{u^2\sin^2\theta}{2g}$

$H_A = \frac{u^2\sin^2\theta}{2g}$

$H_B = \frac{u^2\sin^2\left(90 - \theta\right)}{2g}$

$H_B = \frac{u^2\cos^2\theta}{2g}$

$\frac{H_A}{H_B} = \frac{\sin^2\theta}{\cos^2\theta} - = \tan^2\theta$

(23) Answer : (4)
Solution:
$x = u \times t + \frac{a_x t^2}{2}$
$= 4 \times 3 + \frac{6}{2}(3)^2 = 12 + 27 = 39$ m

(24) Answer : (3)

Solution:
The change in magnitude of velocity is zero, while magnitude of change in velocity is non-zero

(25) **Answer :** (1)
Solution:
$t = \frac{d}{v\cos 0^\circ} = \frac{1}{6} \times 60$
= 10 minute

(26) **Answer :** (2)
Solution:
$T_{max} = 6mg$
$T_{min} = 0$
$T_{max} - T_{min} = 6mg$

(27) **Answer :** (3)
Solution:
$x = (t + 2t^2)$ m
$v = (1 + 4t)$
$a = 4$ m/s^2
$P = \vec{F}\,.\,\vec{v}$
$= 4 \times 4 \times (1 + 4t) = 4 \times 4 \times 5 = 80$ W

(28) **Answer :** (1)
Solution:
Coefficient of restitution
$= \left(\frac{\text{Relative velocity of separation}}{\text{Relative velocity of approach}}\right)$
$e = 1$

(29) **Answer :** (4)
Solution:
$m_1 = \frac{4}{2} = 2$
$4y + bx = 2$
$y = \frac{-bx}{4} + \frac{1}{2}$
$m_2 = \frac{-b}{4}$
$m_1 \times m_2 = -1$
$\frac{-b}{4} \times 2 = -1$
$b = 2$

(30) **Answer :** (3)
Solution:
$W = \vec{F}\,.\,\vec{d}$
$F = kx$
$k_A x_A = k_B x_B$
$\frac{k_A}{k_B} = \frac{x_A}{x_B}$
$\frac{U_A}{U_B} = \frac{\frac{1}{2}k_A\,x_A^2}{\frac{1}{2}k_B\,x_B^2} = \frac{k_A}{k_B}\frac{k_B^2}{k_A^2} = \frac{k_B}{k_A}$
$k_A > k_B$
$U_B > U_A$
$W_A > W_B$

(31) **Answer :** (3)
Solution:
$W = \vec{f}\,.\,\vec{d}$
Work done by conservative force the round trip is zero.

(32) **Answer :** (2)
Solution:
Area under force-displacement graph is work done

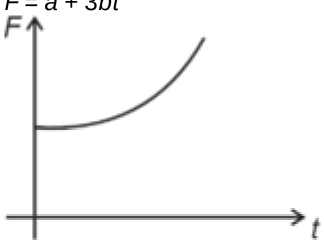
$W = 8 \times 10 + \frac{1}{2} \times 10 \times 8 = 80 + 40 = 120$ J

(33) **Answer :** (4)
Solution:
$mg = kx$
In second case
$3\, mgx_1 = \frac{1}{2} kx_1^2$
$6\, mg = kx_1$
$6kx = kx_1$
$x_1 = 6x$

(34) **Answer :** (2)
Solution:
At lower most position
$\frac{v^2}{l} = 5g$
$T = 6\, mg$
At upper most position
$T = 0$
$\frac{v^2}{l} = g$
At horizontal position
$T = 3\, mg$
Radial acceleration at uppermost position $mg = ma \Rightarrow a = g$
Radial acceleration at lowermost position $T - mg = ma$
$6mg - mg = ma$
$a = 5g$

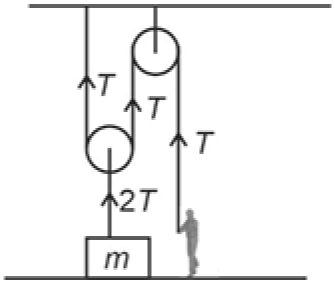
(35) **Answer :** (3)
Solution:
$\overrightarrow{A} \,.\, \overrightarrow{B} = AB\cos\theta$
$6\sqrt{3} = 3 \times 4 \cos\theta$
$\cos\theta = \frac{\sqrt{3}}{2}$
$\theta = \frac{\pi}{6}$ rad

(36) **Answer :** (1)
Solution:
$p = a_0 + at + bt^3$
$F = \frac{dp}{dt}$
$F = \frac{d}{dt}\left(a_0 + at + bt^3\right)$
$F = a + 3bt^2$



(37) **Answer :** (1)
Solution:
$\overrightarrow{P_i} = \overrightarrow{P_t}$
$4M\left(30\hat{i} + 36\hat{j} + 24\hat{k}\right) = M\left(100\hat{i} + 40\hat{j} - 10\hat{k}\right) + 3M\,\overrightarrow{v}$
$\left(120\hat{i} + 144\hat{j} + 96\hat{k}\right) = \left(100\hat{i} + 40\hat{j} - 10\hat{k}\right) + 3\overrightarrow{v}$
$3\overrightarrow{v} = \left(20\hat{i} + 104\hat{j} + 106\hat{k}\right)$
$\overrightarrow{v} = \left(\frac{20}{3}\hat{i} + \frac{104}{3}\hat{j} + \frac{106}{3}\hat{k}\right)$

(38) **Answer :** (3)
Solution:



$2T = mg$
$T = \frac{mg}{2}$

(39) **Answer :** (4)
Solution:
$F = \mu mg$
$4t = 0.4 \times 4 \times 10$
$t = 4s$

(40) **Answer :** (4)
Solution:
The apparent weight of freely falling body is zero

(41) **Answer :** (2)
Solution:
• Frictional force always oppose relative motion between the surface of contact
• Coefficient of static friction is greater than coefficient of kinetic friction for given pair of surfaces
• Friction force is non-conservative force in nature.

(42) **Answer :** (3)
Solution:
$a = \frac{20}{10} = 2$ m/s^2
$N_1 = 5 \times 2$
$N_2 = 2 \times 2$
$\frac{N_1}{N_2} = \frac{5}{2}$

(43) **Answer :** (4)
Solution:
$a = \frac{F}{m_1+m_2} = \frac{40}{5} = 8$ m/s^2
$40 - kx = 2 \times 8$
$x = \frac{24}{400} = 6$ cm

(44) **Answer :** (3)
Solution:
$\mu = \tan\theta = \tan 37° = \frac{3}{4}$

(45) **Answer :** (3)
Solution:
Area under force time graph gives change in linear momentum.

CHEMISTRY

(46) **Answer :** (2)
Solution:

Species	Dipole moment, μ(D)
H_2O	1.85
H_2S	0.95

NH_3	1.47
NF_3	0.23

(47) **Answer :** (3)
Solution:

Species	Bond enthalpy/(kJ mol^{-1})
$O_2(O = O)$	498
$N_2(N \equiv N)$	946
HCl	431
H_2	435.8
HCl	431
H_2	435.8

(48) **Answer :** (1)
Solution:

$K_p = K_c(RT)^{\Delta n_g}$; $\Delta n_g = 1$

$K_c = \frac{1.64}{0.0821 \times 800}$

$K_c = \frac{1.64}{65.68} = 0.025$

$K_c = [CO_2]$

	$CaCO_3(s) \rightleftharpoons$	$CaO(s)$	$+ CO_2(g)$
at t = 0	1 mole	0	0
at $t = t_{eq}$	1 – x	x	x

$K_c = \frac{x}{5}$

x = 5 × 0.025 = 0.125 mole

Amount of CO_2 = 0.125 × 44 = 5.5 g

(49) **Answer :** (3)
Solution:

$2Al + 2NaOH + 6H_2O \rightarrow 2\,Na^+\,[Al(OH)_4]^- + 3H_2$

$AlCl_3$ in acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion

(50) **Answer :** (4)
Solution:

$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$

$B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 6H_2$

$Na_2B_4O_7 + 7H_2O \rightarrow 4H_3BO_3 + 2\,NaOH$

$BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3\,HCl$

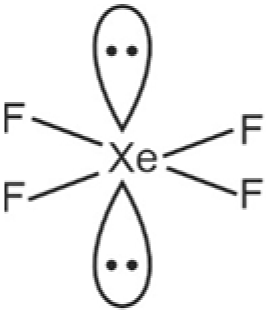
(51) **Answer :** (2)
Solution:

Bond order = $\frac{1}{2}[N_b - N_a]$

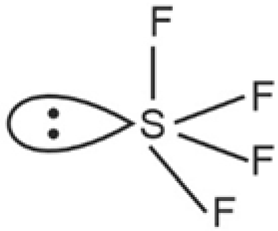
Species	Bond order
N_2^-	2.5
CO	3.0
O_2^{2-}	1.0
F_2^-	0.5

(52) **Answer :** (1)

Solution:



(XeF_4)
(Square planar)



(SF_4)
See-saw

Ethyne (C_2H_2),

$$\underset{sp}{H-C}\equiv\underset{sp}{C-H}$$

(53) **Answer :** (1)
Solution:
If there are more than eight valence electrons around the central atom, termed as the expanded octet

SO_3, PF_5, IF_7, H_3PO_4 and SF_4 have 12, 10, 14, 10 and 10 electrons in their valence shell so they termed as expanded octet.

(54) **Answer :** (3)
Solution:
Increase in volume, reaction will move towards the direction of increase in number of gaseous moles. For the equilibrium

$2\,SO_2(g)+O_2(g)\rightleftharpoons 2\,SO_3(g)$

As volume increases, equilibrium will shift in backward direction.

(55) **Answer :** (4)
Solution:
$K_a \times K_b = K_w = 10^{-14}$

$K_b = \frac{10^{-14}}{10^{-5}} = 10^{-9}$

$pK_b = -\log K_b = 9$

(56) **Answer :** (2)
Solution:

$$\underset{S}{Cd(OH)_2} \rightleftharpoons \underset{S}{Cd^{2+}} + \underset{2S}{2OH^-}$$

$pOH = 4$
$[OH^-] = 10^{-4}$

$2S = 10^{-4}$

$S = 0.5 \times 10^{-4}$

$K_{sp} = [Cd^{2+}]\,[OH^-]^2$

$= 0.5 \times 10^{-4}\,(10^{-4})^2$

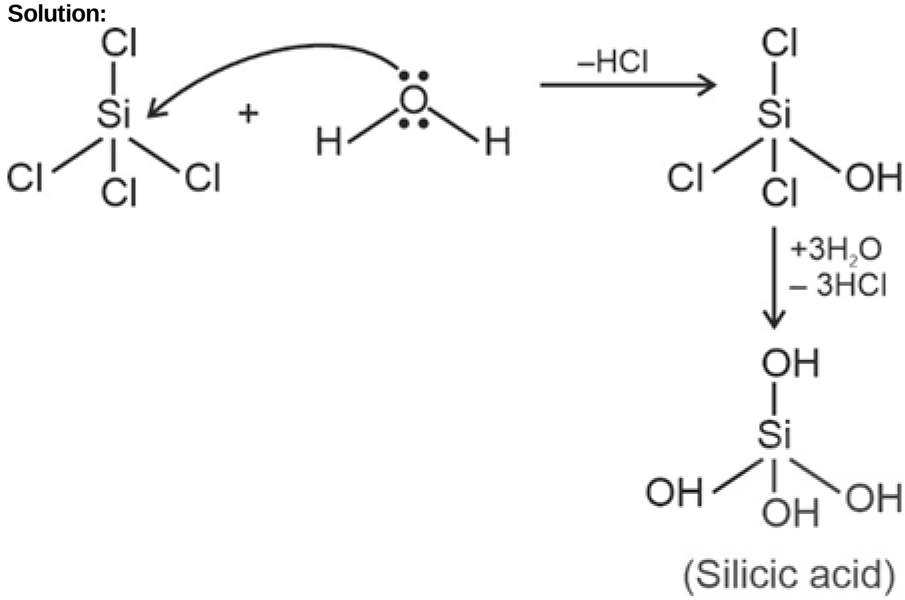
$= 0.5 \times 10^{-12}$

(57) **Answer :** (4)
Solution:

Species	Nature
CO	Neutral
GeO_2	Acidic

Tl_2O_3	Basic
Ga_2O_3	Amphoteric

(58) **Answer :** (1)
Solution:



(59) **Answer :** (3)
Solution:

Bond	Bond enthalpy/kJmol^{-1}
C–C	348
Si–Si	297
Ge–Ge	260
Sn–Sn	240

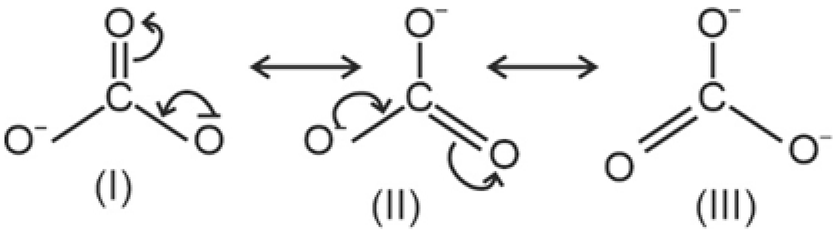
(60) **Answer :** (2)
Solution:

Species	Bond order	Nature
C_2	2.0	Diamagnetic
C_2^+	1.5	Paramagnetic
O_2	2	Paramagnetic
O_2^+	2.5	Paramagnetic
O_2^-	1.5	Paramagnetic
B_2	1	Paramagnetic
B_2^+	0.5	Paramagnetic

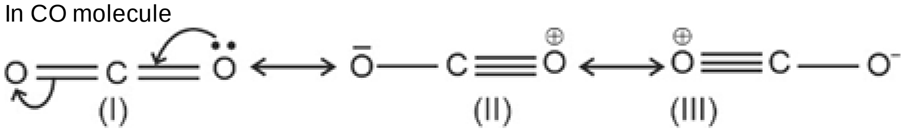
(61) **Answer :** (1)
Solution:

Formal charge on central oxygen atom
$= \left(6 - 2 - \frac{6}{2}\right) = 1$

In CO_3^{2-} ion



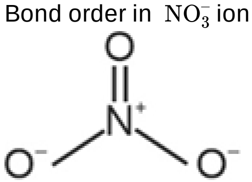
In CO molecule



(62) **Answer :** (2)
Solution:

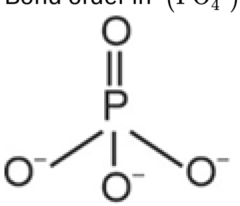
Bond type	Covalent bond length (pm)
O–H	96
C–H	107
N–O	136
C–O	143

(63) **Answer :** (1)
Solution:
Bond order in NO_3^- ion



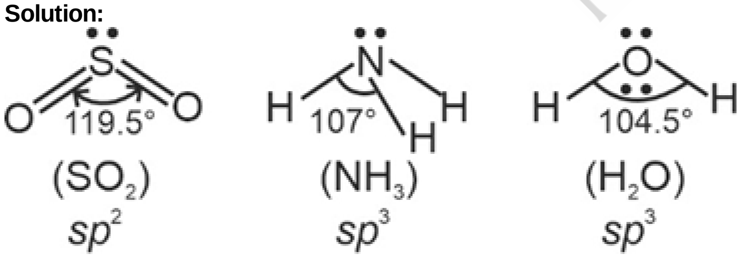
$= \frac{4}{3} = 1.33$

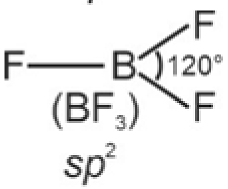
Bond order in $\left(PO_4^{3-}\right)$ ion



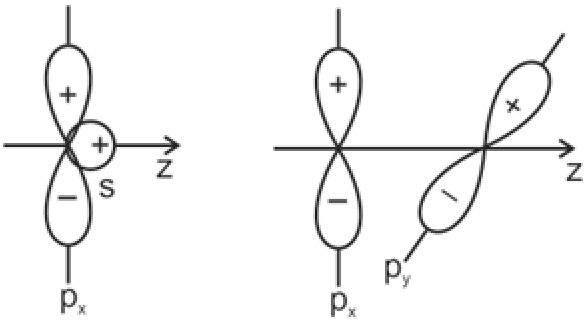
$= \frac{5}{4} = 1.25$

(64) **Answer :** (4)
Solution:





(65) **Answer :** (3)
Solution:

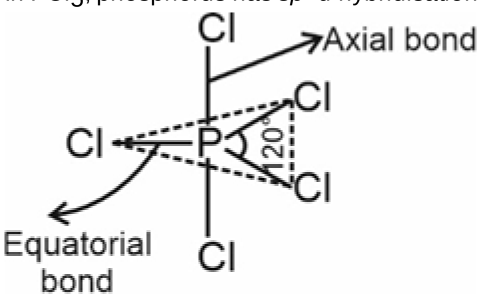


Zero overlapping

(66) Answer : (2)

Solution:

In PCl_5, phosphorus has sp^3d hybridisation



Axial bond pairs suffer more repulsive interaction from the equatorial bond pairs, therefore axial bonds have been found to be slightly longer and hence slightly weaker than the equatorial bonds

(67) Answer : (4)

Solution:

Millimoles of NaOH = 100 × 0.01 = 1mmol

Millimoles of H_2SO_4 = 100 × 0.01 = 1mmol

Millimoles of H^+ ion = 2mmol

So 1 mmol of H^+ react with 1 mmol of OH^- ion

Remaining moles of H^+ = $\frac{1}{1000} = 10^{-3}\,\mathrm{mol}$

$\left[\mathrm{H}^+\right] = \frac{10^{-3}}{200} \times 1000 = 5 \times 10^{-3}$

pH = –log[H^+]

= –log[5 × 10^{-3}] = 3 – 0.6990 = 2.3

(68) Answer : (3)

Solution:

pH of salt of strong base and weak acid is

$\mathrm{pH} = 7 + \frac{1}{2}\left[\mathrm{pK_a} + \log\mathrm{C}\right]$

$= 7 + \frac{1}{2}\left[4.74 + \left(-1.6990\right)\right]$

$= 7 + \frac{1}{2} \times 3.041 = 8.52$

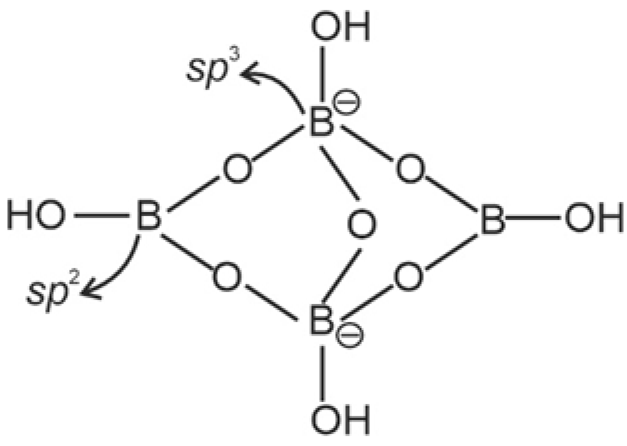
(69) Answer : (2)

Solution:

In exothermic reaction, increase in temperature shifts the equilibrium in the backward direction.
On increasing the concentration of reactant, the equilibrium will shift in the forward direction
On decreasing the concentration of product, the equilibrium will shift in the forward direction

(70) Answer : (4)

Solution:



(Borax)

It has two sp^2 hybridised and two sp^3 hybridised Boron atoms
Boric acid is a weak monobasic Lewis acid because it accept electrons from a hydroxyl ion.

(71) **Answer :** (1)
Solution:

	CH_3COOH	$\rightleftharpoons$ CH_3COO^-	+ H^+
Initial	0.02	–	–
Final	$0.02 - 0.02\alpha$	0.02α	$0.02\alpha + 0.2$

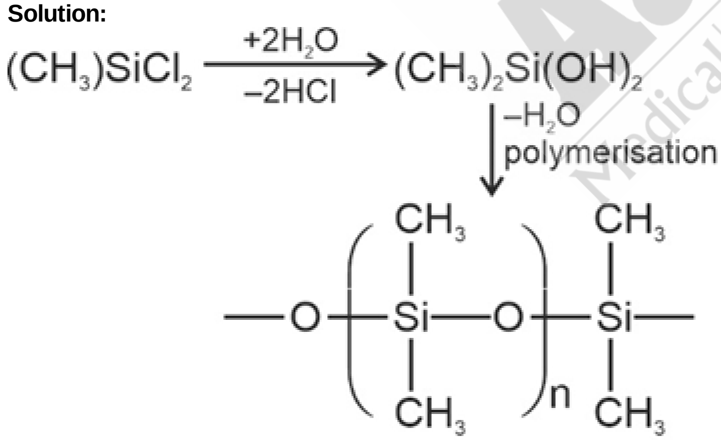
$K_a = \frac{0.02\alpha(0.02\alpha+0.2)}{0.02(1-\alpha)}$

$K_a = \frac{0.2\alpha}{1}$

$\alpha = \frac{1.74\times10^{-5}}{0.2} = 8.7 \times 10^{-5} \times 100 = 8.7 \times 10^{-3} = 0.009\%$

(72) **Answer :** (3)
Solution:
Buckminsterfullerene contains twenty six- membered rings and twelve five-membered rings
Charcoal is obtained by heating wood at high temperatures in the absence of air.

(73) **Answer :** (1)
Solution:

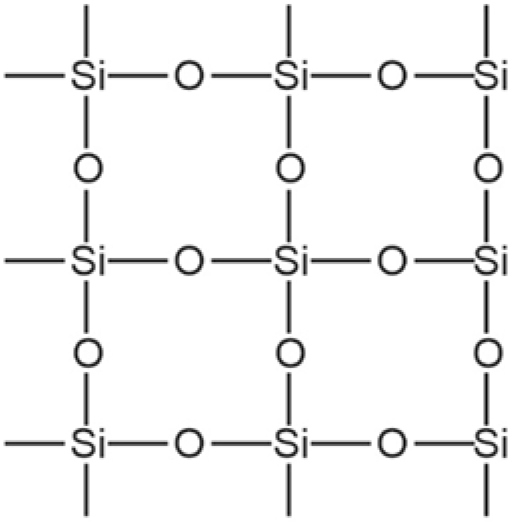


Silicone

$H_3BO_3 \xrightarrow{\Delta} HBO_2 \xrightarrow{\Delta} B_2O_3$

(74) **Answer :** (2)
Solution:
$\Delta G° = -2.303\ RT \log K$
$= -2.303 \times 8.314 \times 300 \log 200$
$= -2.303 \times 8.314 \times 300 \times 2.3010 = -13.22$ kJ/mole

(75) **Answer :** (4)
Solution:



(SiO_2)

SiO_2 is a covalent, three-dimensional network solid in which each silicon atom is covalently bonded in a tetrahedral manner to four oxygen atoms.
ZSM-5 is used to convert alcohols directly into gasoline.

(76) **Answer :** (4)
Solution:

	$HCN \rightleftharpoons$	H^+	$+\ CN^-$
$t = 0$	c	$-$	$-$
$t = t_{eq}$	$c - c\alpha$	$c\alpha$	$c\alpha$

$K_a = \frac{c^2\alpha^2}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$ a << 1

$K_a = c\alpha^2$

$[H^+] = 10^{-4} = c\alpha$

$\alpha = \frac{10^{-4}}{10^{-1}}$

$\alpha = 10^{-3}$

$K_a = 10^{-1} \times (10^{-3})^2$

$K_a = 10^{-7}$

(77) **Answer :** (4)
Solution:
If the volume is kept constant and an inert gas such as argon is added which does not take part in the reaction, the equilibrium remains undisturbed because the addition of an inert gas at constant volume does not change the partial pressure of the reaction.

(78) **Answer :** (1)
Solution:

For acidic buffer

$pH = pK_a + \log\frac{[Salt]}{[acid]}$

$= 4.74 + \log\frac{5}{15}$

= 4.74 + log1 – log3
= 4.74 + 0 –0.4771 = 4.26

(79) **Answer :** (4)
Solution:
Carbon monoxide is a colourless, odourless and almost water insoluble gas

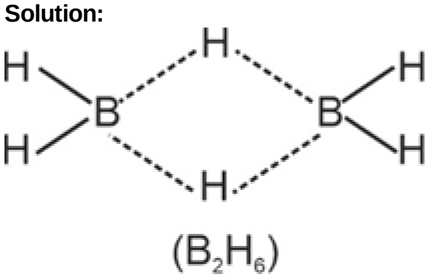
$$\mathrm{HCOOH} \xrightarrow[\text{Conc. } \mathrm{H_2SO_4}]{373\ \mathrm{K}} \mathrm{H_2O} + \mathrm{CO}$$

The highly poisonous nature of CO arises because of its ability to form a complex with haemoglobin which is about 300 times more stable than the oxygen-haemoglobin complex.

(80) Answer : (3)
Solution:

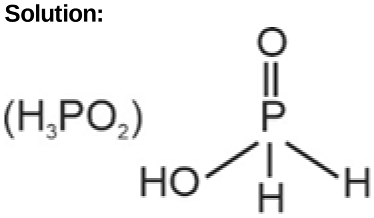
$$\mathrm{K_h} = \frac{10^{-14}}{1.8\times10^{-5}\times1.77\times10^{-5}} = 3.1\times10^{-5}$$

(81) Answer : (3)
Solution:



In B_2H_6, normal 2-centre-2-electron bonds and banana bonds are four and two respectively.

(82) Answer : (2)
Solution:



$H_2PO_2^-$ does not have acidic hydrogen so it does not form conjugate base

Conjugate base of $H_2PO_4^-$ is HPO_4^{2-}

Conjugate base of HPO_4^{2-} is PO_4^{3-}

(83) Answer : (2)
Solution:
Weak base and salt of weak base with strong acid form basic buffer.

It forms basic buffer.

(84) Answer : (1)
Solution:
If $Q_C > K_C$, the reaction will proceed in the backward direction
If $Q_C < K_C$, the reaction will proceed in the forward direction.

(85) Answer : (3)
Solution:

$$\mathrm{AgCl}\left(\mathrm{s}\right) \rightleftharpoons \underset{\mathrm{S}}{\mathrm{Ag^+}}(\mathrm{aq}) + \underset{(\mathrm{S}+0.1)}{\mathrm{Cl^-}(\mathrm{aq})}$$

K_{sp} = (S) (S + 0.1)

$1.8 \times 10^{-10} = S^2 + 0.1S$ (0.1 >> S)

$S = \frac{1.8\times10^{-10}}{0.1} = 18\times10^{-10}$ M = 1.8×10^{-9} M

(86) Answer : (4)
Solution:

$\mathrm{A + 3B \rightleftharpoons 2\,C\ ;\ K_1}$ ...(i)

Reverse the equation (i)

$\mathrm{2C \rightleftharpoons A + 3B}\ ;\ \frac{1}{K_1}$ ...(ii)

Multiply the below equation by 3

$\mathrm{B} + \frac{\mathrm{D}}{2} \rightleftharpoons \mathrm{F}\ ;\ \mathrm{K_3}$

$3B + \frac{3D}{2} \rightleftharpoons 3F\ ;\ \ K_3^3$ ...(iii)

$A + D \rightleftharpoons 2E\ ;\ \ K_2$ ...(iv)

Add eq (ii), (iii) and (iv)

$2C + 3B + \frac{3}{2}D + A + D \rightleftharpoons 3F + 2E + A + 3B$

Then we get

$2C + \frac{5}{2}D \rightleftharpoons 3F + 2E\ ;\ \ K = \frac{K_2 K_3^3}{K_1}$

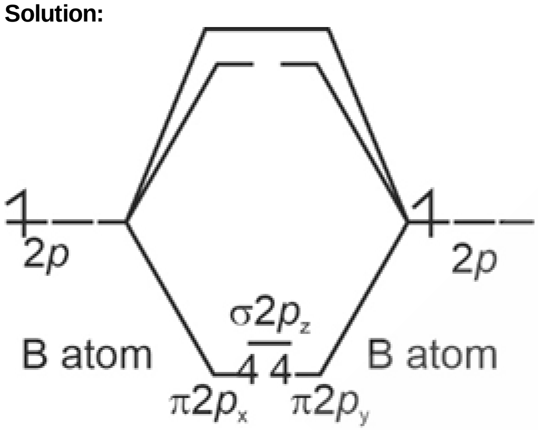
(87) Answer : (2)

Solution:

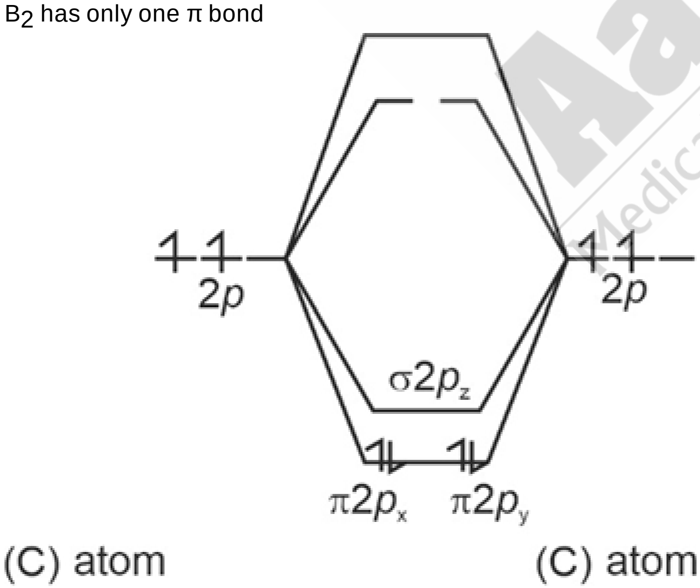
Element	Melting point/(K)
B	2453
Al	933
Ga	303
In	430
Tl	576

(88) Answer : (3)

Solution:



B_2 has only one π bond



C_2 has only 2π bonds

(89) Answer : (1)

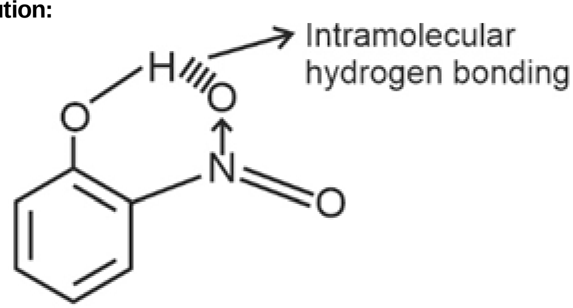
Solution:

For pure water; $[H^+] = [OH^-]$

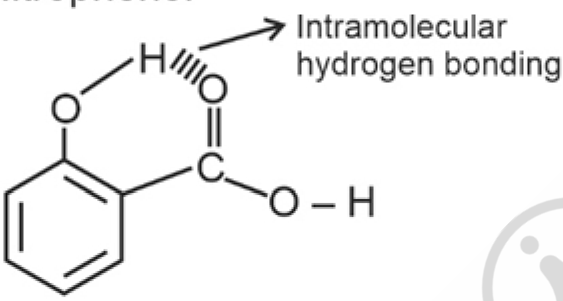
$K_W = [H^+] = [OH^-]$

$10^{-12} = x^2$
$x = 10^{-6}$
$[H^+] = 10^{-6}$
pH = 6

(90) **Answer :** (1)
Solution:



o-Nitrophenol



(Salicylic acid)

In H_2O and NH_3 molecules, intermolecular hydrogen bonding exist

BOTANY

(91) **Answer :** (3)
Solution:
In 1855, Rudolf Virchow first explained that cells divide and new cells are formed from the pre-existing cells (*Omnis cellula-e- cellula*).

(92) **Answer :** (3)
Solution:
Cisternae are found in the endomembrane system of eukaryotic cells. They are generally absent in prokaryotes.

(93) **Answer :** (3)
Solution:
In some bacteria, glycocalyx is present in the form of a thick and tough covering called capsule. It protects the bacterium from the host's immune system.

(94) **Answer :** (4)
Solution:
Bacterial flagella are made up of a protein called flagellin.

(95) **Answer :** (2)
Solution:
Pili are elongated tubular structures, made up of a special protein, pilin. They are involved in the mating process of bacteria.

(96) **Answer :** (3)
Solution:
Integral membrane proteins that run through the lipid bilayer are called trans membrane proteins.
Secondary cell wall is a multi-layered structure. Primary cell wall diminishes with maturity in plants.

(97) **Answer :** (1)

Solution:
Artificial system of classification was proposed by Aristotle and Linnaeus.

(98) Answer : (4)
Solution:
Chemotaxonomy is based on the chemical constituent of the plants. It uses DNA sequencing,chemical nature of proteins etc. to resolve confusion in classification.

(99) Answer : (2)
Solution:
Anisogamy involves the fusion of two gametes, which are dissimilar in size. It is observed in *Eudorina*.

(100) Answer : (3)
Solution:
Phaeophyceae or brown algae have 2 laterally attached unequal flagella in zoospores.

(101) Answer : (2)
Solution:
Red algae is found at the great depths in oceans.
Algin is a hydrocolloid found in brown algae.

(102) Answer : (2)
Solution:
Asexual mode of reproduction is observed in protistans, fungi and plants also.

(103) Answer : (3)
Solution:
Nitrosomonas and *Nitrobacter* are the examples of eubacteria, which show chemosynthetic mode of nutrition.
Introns are present in the genetic sequence of archaebacteria but not in other bacteria.

(104) Answer : (1)
Solution:
Euglenoids show autotrophic mode of nutrition during day time. However, in the absence of light, they show heterotrophic mode of nutrition.

(105) Answer : (2)
Solution:
In dinoflagellates, cell wall has stiff cellulosic plates on the outer surface.

(106) Answer : (3)
Solution:
Autolysis or autophagy is associated with lysosomes.

(107) Answer : (4)
Solution:
Synthesis of carbohydrates occurs in chloroplast but not in mitochondria.

(108) Answer : (2)
Solution:
(1) Luminal compartment — Endoplasmic reticulum
(2) SER — Ca^{2+} release in muscle cells
(3) Acrosome — Modified Golgi apparatus
(4) Residual bodies — Formed by lysosomes

(109) Answer : (2)
Solution:
Bryophytes are known as amphibians of the plant kingdom.

(110) Answer : (3)
Solution:
Multicellular and branched rhizoids are absent in liverworts.

(111) Answer : (3)
Solution:
Salvinia is a heterosporous pteridophyte.

(112) Answer : (3)
Solution:
Amoeba does not possess cilia.

(113) Answer : (4)

Solution:
Albugo belongs to the class Phycomycetes. The members of Phycomycetes have aseptate mycelium.

(114) Answer : (2)
Solution:
Prothallus of pteridophytes is multicellular, inconspicuous, photosynthetic and thalloid.

(115) Answer : (4)
Solution:
Bryophytes are the first embryophytes and pteridophytes are the seedless vascular plants. Both bryophytes and pteridophytes need water for fertilisation and therefore, they have flagellated antherozoids. Both prevent soil erosion and majority of the pteridophytes are homosporous.

(116) Answer : (4)
Solution:
Amyloplasts are starch-containing colourless plastids.

(117) Answer : (3)
Solution:
Microfilaments play an important role in the formation of pseudopodia.

(118) Answer : (2)
Solution:
Living cells were first observed by Anton von Leeuwenhoek under his own designed microscope.

(119) Answer : (3)
Solution:
Fungal cell wall is mainly composed of chitin.

(120) Answer : (3)
Solution:
Arms in both sub-metacentric and acrocentric chromosomes are referred to as p-arms and q-arms

(121) Answer : (2)
Solution:
Few chromosomes have non-staining secondary constrictions at certain location. This gives the appearance of a small fragment called the satellite.

(122) Answer : (2)
Solution:
Centrioles are known as diplosomes. They have 9 + 0 arrangement of microtubules. The centre of the centriole possesses a rod-shaped proteinaceous mass known as hub.

(123) Answer : (2)
Solution:
Na^+/K^+ pump is an example of active transport.

(124) Answer : (1)
Solution:
Lysosomes is a polymorphic cell organelle.

(125) Answer : (4)
Solution:
(i) Chromatin is named so because it gets stained with certain basic dyes.
(ii) Chromatin is mainly composed of DNA and basic proteins called histones. However, it also contains RNA and some non-histone proteins.

(126) Answer : (1)
Solution:
Protonema is a green branched and filamentous stage produced by the spores of mosses.

(127) Answer : (2)
Solution:
Selaginella belongs to the class, Lycopsida and *Equisetum* belongs to the class Sphenopsida.

(128) Answer : (3)
Solution:
Sphagnum is a moss and it reproduces asexually by means of fragmentation and budding in secondary protonema.

(129) Answer : (1)

Solution:
In gymnosperms, the male and female gametophytes do not have a free-living existence.

(130) Answer : (2)
Solution:
In conifers, leaves are needle-like with thick cuticle.

(131) Answer : (2)
Solution:
M. W. Beijerinek demonstrated that the extract of infected plant of tobacco could cause infection in a healthy plant.

(132) Answer : (3)
Solution:
Bladderwort and the venus fly trap are the examples of insectivorous plants, whereas *Cuscuta* is a parasitic plant.

(133) Answer : (3)
Solution:
Secondary mycelium is observed in the members of Basidiomycetes. *Alternaria solani* belongs to the class Deuteromycetes.

(134) Answer : (1)
Solution:
Viroids are infectious RNA particles, devoid of a protein coat.

(135) Answer : (2)
Solution:
Eukaryotic cells have 80S ribosome and cell organelles. *Amoeba proteus* is an example of a single celled eukaryotic organism.

ZOOLOGY

(136) Answer : (4)
Solution:
All fishes have a streamlined body. This shape helps them to swim more efficiently by reducing water resistance.
Cyclostomes are marine but migrate for spawning to fresh water.
Ichthyophis is a limbless amphibian.
Reptiles are mostly terrestrial animals. Some reptiles (turtles) live in water.

(137) Answer : (4)
Solution:
In platyhelminths, organ -level of organisation is seen. Porifers have cellular level of organisation. Coelenterates and ctenophores have tissue level of body organisation.

(138) Answer : (1)
Solution:
Ancylostoma (Hookworm) belongs to the phylum Aschelminthes. Other members of this phylum are *Ascaris* (Roundworm), *Wuchereria* (Filaria worm)

Phylum	Examples
Annelida	*Nereis*, *Pheretima* (Earthworm) and *Hirudinaria* (Blood sucking leech)
Platyhelminthes	*Taenia* (Tapeworm), *Fasciola* (Liver fluke)
Ctenophora	*Pleurobrachia, Ctenoplana*

(139) Answer : (4)
Solution:
Nereis belongs to the phylum Annelida. Aquatic annelids like *Nereis* possess lateral appendages, parapodia, which help in swimming. It is dioecious.

(140) Answer : (4)
Solution:
The most distinctive feature of echinoderms is the presence of water vascular system. *Asterias* (Star fish), *Echinus* (Sea urchin), *Antedon* (Sea lily) are echinoderms. *Aplysia* (Sea hare) is a mollusc.

(141) Answer : (4)
Solution:

Lipids are generally water-insoluble.
Trihydroxy propane is a simple lipid.
A triglyceride is composed of one glycerol molecule and three fatty acids.

(142) Answer : (1)
Solution:
In amphibians (*Hyla*), sexes are separate; fertilisation is mostly external. They are oviparous and development is indirect.
In reptiles (*Hemidactylus*), sexes are separate; fertilisation is internal. They are oviparous and exhibit direct development.
In birds (*Aptenodytes*) sexes are separate; fertilisation is internal. They are oviparous and show direct development.
In mammals (*Balaenoptera*), sexes are separate; fertilisation is internal. They are mostly viviparous with few exceptions and show direct development.

(143) Answer : (1)
Solution:
In members belonging to the class Chondrichthyes, mouth is located ventrally. Gill slits are separate and without operculum (gill cover). The skin is tough, containing minute placoid scales. Due to absence of air bladder, they have to swim constantly to avoid sinking. *Exocoetus, Hippocampus* and *Clarias* belong to the class Osteichthyes. They have cycloid/ctenoid scales. They also possess operculum and air bladder.

(144) Answer : (1)
Solution:
In the members of the subphylum Urochordata, notochord is present only in its larval tail. For example- *Ascidia, Salpa* and *Doliolum*.
In cephalochordates, notochord extends from head to tail region and is persistent throughout their life. For example- *Branchiostoma.*
In cyclostomes, notochord is persistent throughout the life. *Petromyzon (*Lamprey) and *Myxine* (Hagfish) are cyclostomes.

(145) Answer : (2)
Solution:
Torpedo is a marine cartilaginous fish which has electric organs.*Trygon* has poison sting. *Pristis* (Saw fish) and *Carcharodon* (Great white shark) are marine cartilaginous fishes but they do not have electric organs.

(146) Answer : (1)
Solution:
Compound epithelium covers the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.
In fallopian tubes, simple ciliated columnar epithelium is present.
In alveoli, simple squamous epithelium is present.

(147) Answer : (4)
Solution:
There are two types of loose connective tissues ;(i) Areolar tissue (ii) Adipose tissue.
Areolar tissue serves as a support framework for epithelium. It is present beneath the skin.
Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently.

(148) Answer : (4)
Solution:
Neurons, the unit of neural system, are excitable cells. The neuroglial cells which constitute the rest of the neural system protect and support the neurons.

(149) Answer : (1)
Solution:
In the dense regular connective tissues, the collagen fibres are present in rows between the many parallel bundles of fibres. Ligament (attaches one bone to another bone) is an example of dense regular connective tissue.

(150) Answer : (2)
Solution:
Tight junctions help to stop substances from leaking across a tissue. Adhering junctions perform cementing to keep neighbouring cells together. Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules. Communication junctions are present in cardiac muscle tissue.

(151) Answer : (2)
Solution:
Body is covered by a calcerous shell and is unsegmented with a distinct head, muscular foot and visceral hump

(152) Answer : (3)
Solution:
Reproduction takes place only by sexual means. Fertilisation is external with indirect development

(153) Answer : (1)
Solution:

Element	% Weight of	
	Earth's crust	**Human body**
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

(154) Answer : (4)
Solution:
Most of the enzymes are proteins. *e.g.* trypsin.
Proteins are made up of different amino acids, thus they are heteropolymers.
Some of them are receptors.
They are biomacromolecules.

(155) Answer : (4)
Solution:
Two polynucleotide chains of B-DNA are wound around the same axis and are held together by complementary base pairing between nitrogenous bases of two strands in the same plane. Adenine of one strand forms two hydrogen bonds with thymine of other strand and guanine forms three hydrogen bonds with cytosine.

(156) Answer : (2)
Solution:
Exoskeleton of arthropods is composed of a complex polysaccharide called chitin. Chitin is a homopolymer.
The monomeric unit of chitin is N-acetyl glucosamine

(157) Answer : (3)
Solution:
The long protein chain is folded upon itself like a hollow woollen ball, giving rise to the tertiary structure of protein. Tertiary structure gives us a 3-dimensional view of a protein. Adult human haemoglobin consists of 4 subunits. Two of these are identical to each other. Since, two subunits of α-type and two subunits of β-type together constitute the human haemoglobin (Hb), thus, it is called a tetramer. Proline is a heterocyclic amino acid and valine is a neutral amino acid. Cysteine and methonine are sulphur containing amino acids

(158) Answer : (3)
Solution:
The chemical and physical properties of amino acids are essentially of the amino, carboxyl and the 'R' functional groups. Based on the number of amino groups and carboxyl groups, there are acidic (*e.g.* glutamic acid and aspartic acid), basic (lysine) and neutral (valine and glycine) amino acids

(159) Answer : (2)
Solution:
Transferases (class II): Enzymes catalysing a transfer of a group, G(other than hydrogen) between a pair of substrate S and S′ *e.g.*
$S - G + S' \rightarrow S + S' - G$
Class I → Oxidoreductases/dehydrogenases
Class III → Hydrolases
Class IV → Lyases

(160) Answer : (1)
Solution:
Starch forms helical secondary structures. Thus, it can hold I_2 molecules in its helical portion. The starch-I_2 is blue in colour. Cellulose does not contain complex helices and hence cannot hold I_2 molecules.

(161) Answer : (4)
Solution:

Glucose – Hexose sugar that does not contain any free carboxyl group; but it does contain a carbonyl group that can be oxidized to a carboxylic acid.
Cholesterol – Contains four homocylic rings
Lecithin – Phospholipid; lipids are not polymeric in nature

(162) Answer : (2)
Solution:
Fructose and adenine will be found in the acid-soluble fraction.
The acid-insoluble fraction has only four types of organic compounds *i.e.,* proteins (Collagen), nucleic acids, polysaccharides (Cellulose and glycogen) and lipids.

(163) Answer : (2)
Solution:

Pigments	Carotenoids, Anthocyanins, *etc*.
Alkaloids	Morphine, Codeine, *etc*
Terpenoides	Monoterpenes, Diterpenes, *etc*
Essential oils	Lemon grass oil, *etc*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc*
Polymeric substances	Rubber, gums, cellulose

(164) Answer : (4)
Solution:
The protein part of an enzyme, that joins with any co-factor to make the enzyme catalytically active, is called the apoenzyme.

(165) Answer : (2)
Solution:
External ear is absent in frogs and tympanum is seen externally. On land, the buccal cavity, skin and lungs act as the respiratory organs

(166) Answer : (1)
Solution:
In frogs, both neural system and endocrine system are present. Frogs serve an important link of food chain and food web in the ecosystem

(167) Answer : (2)
Solution:
In both male and female cockroaches, the 10^{th} segment bears a pair of jointed filamentous structures called anal cerci. Only male cockroach bear a pair of short, thread-like anal style.

(168) Answer : (3)
Solution:
Brain of frog is divided into fore-brain, mid-brain and hind-brain. Fore-brain includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon. Mid-brain is characterised by a pair optic lobes. Hind-brain consists of cerebellum and medulla oblongata.

(169) Answer : (1)
Solution:
In cockroaches, compound eyes are situated at the dorsal surface of the head. Each eye consists of about 2000 hexagonal ommatidia. With the help of several ommatidia, a cockroach can receive several images of an object. This kind of vision is known as mosaic vision with more sensitivity but less resolution, being common during night (hence called nocturnal vision).

(170) Answer : (3)
Solution:
In both male and female frogs, body is divisible into head and trunk. In both sexes, hind limbs end in five digits. Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the first digit of the fore limb which are absent in female frogs.

(171) Answer : (2)
Solution:

In the given chain, hydrophobic bonds, disulphide bond (Cys-Cys) and other non-covalent bonds are operating to stabilise the structure.
A protein is imagined as a line, where the left end is represented by the first amino acid and the right end is represented by the last amino acid. The first amino acid is also called the N-terminal amino acid. The last amino acid is called the C-terminal amino acid.
In the given structure, there are 22 amino acids. Tyrosine is an aromatic amino acid.

(172) Answer : (2)
Solution:
Most of the members of the phylum Platyhelminthes are endoparasites found in animals including human beings. Hooks and suckers are present in the parasitic forms.

(173) Answer : (1)
Solution:
Taenia belongs to the phylum Platyhelminthes. They have dorso-ventrally flattened body. Segmentation is absent. Segmentation is present in annelids, arthropods and chordates. Aschelminths have a complete alimentary canal with well-developed muscular pharynx (*e.g. Ascaris* and *Wuchereria*). *Hirudinaria* belongs to the phylum Annelida. Aquatic forms have parapodia for locomotion. *Locusta* belongs to the phylum Arthropoda; they have jointed appendages.

(174) Answer : (1)
Solution:
In echinoderms, an excretory system is absent. In hemichordates, proboscis gland acts as the excretory organ. In molluscs, feather-like gills are present in the mantle cavity that have respiratory and excretory functions. In platyhelminths, flame cells help in osmoregulation and excretion.

(175) Answer : (2)
Solution:
In cyclostomes, cranium and vertebral column are cartilaginous. In cartilaginous fishes, the notochord is persistent throughout the life. Urochordates have an open circulatory system, so all chordates do not have a closed circulatory system. In chordates, heart is located ventrally.

(176) Answer : (2)
Solution:
The male reproductive system consists of a pair of testes one lying on each lateral side in the 4^{th}-6^{th} abdominal segments. In females, ovaries lie laterally in the 2^{nd}-6^{th} abdominal segments. Mushroom-shaped gland is present in the 6^{th}-7^{th} abdominal segments which function as an accessory reproductive gland in males . A pair of spermathecae is present in the 6^{th} segment of females which open into the genital chamber.

(177) Answer : (1)
Solution:
Special venous connection between liver and intestine as well as the kidneys and lower parts of the body are present in frogs. The former is called hepatic portal system and the latter is called the renal portal system. Hepatic portal system is also present in humans. Coronary circulation is present in the heart, hypophyseal portal system is present between hypothalamus and anterior pituitary. Renal portal system is absent in humans.

(178) Answer : (3)
Solution:
The thin walled urinary bladder is present ventral to the rectum which also opens in the cloaca. The ureters open directly into cloaca.

(179) Answer : (4)
Solution:
Stability is something related to the energy status of the molecule or the structure. Transition state has the highest energy level or it is the most unstable state. Enzyme-substrate complex represents the transition state.

(180) Answer : (4)
Solution:
When nitrogenous bases are attached to sugar, they are called nucleosides. Examples: adenosine, guanosine, thymidine, uridine and cytidine.